

Math 526, F08
Quiz 7

Name: *Answer*
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Are these pairs of vectors orthonormal or only orthogonal or only independent?

(a) $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$

linearly independent

(b) $\begin{bmatrix} -0.6 \\ 0.8 \end{bmatrix}, \begin{bmatrix} 0.4 \\ 0.3 \end{bmatrix}.$

orthogonal

(c) $\begin{bmatrix} \frac{1}{2} \cos \theta \\ -\frac{1}{2} \sin \theta \\ \frac{\sqrt{3}}{2} \end{bmatrix}, \begin{bmatrix} \sin \theta \\ \cos \theta \\ 0 \end{bmatrix}.$

orthonormal.

Problem 2. (10 points) Find an orthonormal basis q_1, q_2, q_3 for the column space of

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 5 \\ 0 & 3 & 6 \end{bmatrix} \quad \text{using the Gram-Schmidt process.}$$

$\downarrow \quad \downarrow \quad \downarrow$
 $a_1 \quad a_2 \quad a_3$

Solution: $\tilde{q}_1 = a_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \|\tilde{q}_1\|^2 = 1$

$$\begin{aligned} \tilde{q}_2 &= a_2 - \frac{\tilde{q}_1^T a_2}{\tilde{q}_1^T \tilde{q}_1} \tilde{q}_1 = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} - \left(\frac{2}{1}\right) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}, \quad \|\tilde{q}_2\|^2 = 9 \end{aligned}$$

$$\begin{aligned} \tilde{q}_3 &= a_3 - \frac{\tilde{q}_1^T a_3}{\tilde{q}_1^T \tilde{q}_1} \tilde{q}_1 - \frac{\tilde{q}_2^T a_3}{\tilde{q}_2^T \tilde{q}_2} \tilde{q}_2 \\ &= \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} - \left(\frac{4}{1}\right) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \left(\frac{18}{9}\right) \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix}, \quad \|\tilde{q}_3\|^2 = 25 \end{aligned}$$

$$\Rightarrow q_1 = \frac{\tilde{q}_1}{\|\tilde{q}_1\|} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad q_2 = \frac{\tilde{q}_2}{\|\tilde{q}_2\|} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad q_3 = \frac{\tilde{q}_3}{\|\tilde{q}_3\|} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$